CLINICAL STUDY REPORT

A Phase 2 Study of Kelvaparib in Recurrent High-grade Serous Ovarian Carcinoma

Study KLV-7704

Investigational product	Kelvaparib (KLV-3308), oral PARP1-selective inhibitor
Indication studied	High-grade serous ovarian carcinoma
Development phase	Phase 2
Sponsor	Marnholt Oncology GmbH
Protocol number	KLV-7704
First subject enrolled	8 June 2018
Data cut-off date	30 April 2020
Database lock	28 May 2020
Report date	12 November 2020
Report status	Final
Good Clinical Practice	Conducted in accordance with the Declaration of Helsinki and ICH Good Clinical Practice.

2 SYNOPSIS

Name of sponsor	Marnholt Oncology GmbH
Name of finished product	KLV-3308
Name of active ingredient	Kelvaparib (KLV-3308)
Title of study	A Phase 2 Study of Kelvaparib in Recurrent High-grade Serous Ovarian Carcinoma
Study number	KLV-7704
Phase of development	Phase 2
Studied period	8 June 2018 to 30 April 2020
Analysis	Final analysis
Objectives	Primary: objective response rate by investigator assessment. Secondary: duration of response, progression-free survival, overall survival and the incidence and severity of treatment-emergent adverse events, and pharmacokinetics.
Methodology	An open-label, multicentre, single-arm study. Approximately 81 subjects receive kelvaparib 300 mg orally twice daily on a continuous schedule.
Number of subjects	81 subjects received at least one dose of kelvaparib.
Diagnosis and main criteria for inclusion	Adults with histologically or cytologically confirmed recurrent high-grade serous ovarian carcinoma, platinum-sensitive or platinum-resistant, measurable disease assessable by RECIST version 1.1, ECOG performance status 0 to 2, adequate organ function.
Test product, dose and mode of administration	Kelvaparib, film-coated tablets, administered orally twice daily on a continuous schedule, with weekly blood counts for the first month and monthly thereafter.
Duration of treatment	Until disease progression, unacceptable toxicity or withdrawal of consent. Median duration 105 days.
Criteria for evaluation - efficacy	Best overall response by RECIST version 1.1, and duration of response, progression-free survival, overall survival.
Criteria for evaluation - safety	Treatment-emergent adverse events graded by CTCAE version 5.0, laboratory parameters and vital signs.
Statistical methods	The primary endpoint is summarised with an exact Clopper-Pearson 95 per cent confidence interval. The primary objective is met if the lower limit excludes the prespecified threshold of 15 per cent, which is the rate reported for this population with the therapy ordinarily offered. A sample size of 81 provides at least 90 per cent power at a one-sided significance level of 0.025 under an assumed true response rate of 31 per cent.
Summary - efficacy	The overall response rate was 33 of 80 subjects (41.2 per cent, 95 per cent CI (30.4, 52.8)). Complete response in 9 subjects (11.2 per cent). Median progression-free survival 6.7 months.
Summary - safety	Grade 3 or higher events in 30 subjects (37.0 per cent), serious events in 17 (21.0 per cent). Nausea was the most frequent event (56 subjects, 69.1 per cent). Myelodysplastic syndrome or acute myeloid leukaemia, an event of special interest, occurred in 3 subjects (3.7 per cent). 8 discontinued because of an adverse event; 31 deaths were reported.

Conclusion	Kelvaparib produced antitumor activity in recurrent high-grade serous ovarian carcinoma, platinum-sensitive or platinum-resistant, with an overall response rate of 41.2 per cent. The primary objective was met: the lower limit of the two-sided 95 per cent confidence interval (30.4 per cent) excludes the prespecified threshold of 15 per cent. The safety profile was consistent with that established in earlier studies and no new signal was identified.
Date of report	12 November 2020

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4 LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation	Definition
ALT	alanine aminotransferase
AST	aspartate aminotransferase
AUC0-12	area under the concentration-time curve over one dosing interval (0 to 12 hours)
AUC0-24	area under the concentration-time curve from 0 to 24 hours
BRCA	breast cancer susceptibility gene
BRCA1	breast cancer susceptibility gene 1
BRCA2	breast cancer susceptibility gene 2
CA-125	cancer antigen 125
CI	confidence interval
CL/F	apparent oral clearance
Cmax	maximum plasma concentration
CR	complete response
CTCAE	Common Terminology Criteria for Adverse Events
Ctrough	trough plasma concentration
CV	coefficient of variation
ECOG	Eastern Cooperative Oncology Group
ICH	International Council for Harmonisation
KLV-3308	kelvaparib (compound code)
MedDRA	Medical Dictionary for Regulatory Activities
NE	not estimable
PFS	progression-free survival
PR	partial response
Q1	first quartile
Q3	third quartile
RECIST	Response Evaluation Criteria in Solid Tumours
SD	stable disease
t1/2	terminal elimination half-life
TEAE	treatment-emergent adverse event
tmax	time to maximum plasma concentration

5 ETHICS

5.1 Independent ethics committee or institutional review board

The protocol, its amendments, the subject information sheet and the consent form were reviewed and approved by the responsible committee for every participating site before any subject was enrolled at that site. Approvals are held in the trial master file.

5.2 Ethical conduct of the study

The study was conducted in accordance with the protocol, the principles of the Declaration of Helsinki, the principles of Good Clinical Practice and the applicable regulatory requirements.

5.3 Subject information and consent

Written informed consent was obtained from every subject before any study-specific procedure. Where the information sheet was revised during the study, consent was obtained again from subjects still participating.

6 INVESTIGATORS AND STUDY ADMINISTRATIVE STRUCTURE

The study was conducted at the sites listed in Appendix 16.1.4, identified by number. Investigator names are not recorded in this report. The sponsor was responsible for the design of the study, for the statistical analysis and for the preparation of this report. Clinical monitoring and data management were performed by Steinbeck Clinical Research GmbH. Laboratory analyses were performed at a central laboratory. Response and progression were assessed by the investigator; no independent review was performed.

7 INTRODUCTION

This report describes study KLV-7704, A Phase 2 Study of Kelvaparib in Recurrent High-grade Serous Ovarian Carcinoma. The study was conducted to evaluate Kelvaparib in subjects with recurrent high-grade serous ovarian carcinoma, platinum-sensitive or platinum-resistant. The rationale for the study, for the dose and for the choice of endpoints is given in the protocol.

8 STUDY OBJECTIVES

Table 8-1 Study objectives

Type	Objective
Primary	Objective response rate by investigator assessment
Safety	To characterise the safety and tolerability of Kelvaparib.

9 INVESTIGATIONAL PLAN

9.1 Overall study design and plan

An open-label, multicentre, single-arm study. Approximately 81 subjects receive kelvaparib 300 mg orally twice daily on a continuous schedule.

9.2 Discussion of study design, including choice of control

The design was appropriate to the question. A single-arm design was used because the primary endpoint is a within-subject response interpretable against a prespecified threshold. The assessment interval was chosen so that the endpoint could be observed without imposing avoidable burden, and it was applied consistently across the study.

9.3 Selection of study population

Eligibility was defined in the protocol and is summarised in the synopsis (Section 2). Every subject was required to meet all inclusion criteria and none of the exclusion criteria; no waivers were granted.

9.4 Treatments

Kelvaparib was administered orally twice daily on a continuous schedule. Dose modification followed the rules in the protocol. Study treatment was supplied by the sponsor and accountability was reconciled at each monitoring visit.

9.5 Efficacy and safety variables

Efficacy was assessed according to RECIST version 1.1. Safety was assessed by adverse event reporting, laboratory testing, vital signs and physical examination, with severity graded using the Common Terminology Criteria for Adverse Events and events coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 23.1; concomitant medications were coded using the WHO Drug Dictionary (WHODrug) Global B2, September 2020. Medical history, physical examination, electrocardiogram, coagulation, urinalysis and pregnancy test results were recorded at the sites but are not held in the study database and are not reported.

9.6 Data quality assurance

Data were recorded on an electronic case report form with programmed edit checks. Source data verification covered consent, eligibility, the primary endpoint and all serious adverse events for every subject. Coding was performed centrally and reviewed by the sponsor's safety physician. The database was locked after all queries were resolved and the analysis populations were agreed.

9.7 Statistical methods planned in the protocol

The primary endpoint is summarised with an exact Clopper-Pearson 95 per cent confidence interval. The primary objective is met if the lower limit excludes the prespecified threshold of 15 per cent, which is the rate reported for this population with the therapy ordinarily offered. A sample size of 81 provides at least 90 per cent power at a one-sided significance level of 0.025 under an assumed true response rate of 31 per cent.

9.8 Changes in the conduct of the study or the planned analyses

The protocol was first approved as version 1.0 on 25 May 2018. It was amended twice before the data cut-off of 30 April 2020:

Amendment 1 (protocol version 2.0, 31 January 2019): Clarification of eligibility and monitoring. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. No change to the analysis populations or to the primary endpoint is introduced by this amendment.

Amendment 2 (protocol version 3.0, 18 June 2019): Revision following emerging safety data. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. No change to the analysis populations or to the primary endpoint is introduced by this amendment.

The protocol of record for this report is version 3.0 (amendment 2, 18 June 2019).

Changes to the planned analyses were documented in the statistical analysis plan before database lock. No analysis reported here was devised after the data were examined.

10 STUDY SUBJECTS

10.1 Disposition of subjects

81 subjects were enrolled and treated. Disposition is in Table 10-1. Of these, 61 discontinued study treatment, most commonly for physician decision.

Table 10-1 Subject Disposition – Safety Analysis Set

Disposition	Kelvaparib 300 mg (N=81)	Total (N=81)
Subjects treated, n	81	81
Treatment ongoing at data cut-off, n (%)	20 (24.7)	20 (24.7)
Discontinued study treatment, n (%)	61 (75.3)	61 (75.3)
Disease progression	15 (18.5)	15 (18.5)
Adverse event	4 (4.9)	4 (4.9)
Death	5 (6.2)	5 (6.2)
Withdrawal by subject	11 (13.6)	11 (13.6)
Physician decision	17 (21.0)	17 (21.0)
Lost to follow-up	7 (8.6)	7 (8.6)
Protocol deviation	2 (2.5)	2 (2.5)

Percentages are based on the number of subjects treated within each group.

10.1.1 Reasons for discontinuation

Physician decision. Recorded for 17 subjects (21.0 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Disease progression. Recorded for 15 subjects (18.5 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Withdrawal by subject. Recorded for 11 subjects (13.6 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Lost to follow-up. Recorded for 7 subjects (8.6 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Death. Recorded for 5 subjects (6.2 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Adverse event. Recorded for 4 subjects (4.9 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Protocol deviation. Recorded for 2 subjects (2.5 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

10.2 Protocol deviations

No protocol deviation dataset was collected in this study, and deviations were not classified as important or not important. The deviations held in the data are those recorded as the reason for discontinuing study treatment: 2 subjects (2.5 per cent of those treated, Table 10-1). These subjects are listed at Appendix 16.2.2. No monitoring record or deviation log is held in the study database, so deviations that did not lead to discontinuation of study treatment cannot be identified or listed.

11 EFFICACY EVALUATION

11.1 Data sets analysed

Table 11-1 Analysis sets

Analysis set	Subjects	Definition
Enrolled	81	All subjects assigned to treatment.
Safety	81	All subjects who received at least one dose.
Full analysis	80	All subjects in the safety analysis set with at least one post-baseline disease assessment.
Efficacy evaluable	80	Subjects with at least one evaluable post-baseline assessment.

11.2 Demographic and other baseline characteristics

Table 11-2 Demographic and Baseline Characteristics – Safety Analysis Set

Characteristic	Kelvaparib 300 mg (N=81)	Total (N=81)
Age (years)		
Mean (SD)	64.9 (8.83)	64.9 (8.83)
Median	65.0	65.0
Min, max	38, 88	38, 88
>= 65 years, n (%)	41 (50.6)	41 (50.6)
Sex, n (%)		
Male	0 (0.0)	0 (0.0)
Female	81 (100.0)	81 (100.0)
Race, n (%)		
White	70 (86.4)	70 (86.4)
Asian	6 (7.4)	6 (7.4)
Black or African American	3 (3.7)	3 (3.7)
Other	2 (2.5)	2 (2.5)
ECOG performance status, n (%)		
0	26 (32.1)	26 (32.1)
1	47 (58.0)	47 (58.0)
2	8 (9.9)	8 (9.9)
BRCA mutation status, n (%)		
BRCA1 mutated	19 (23.5)	19 (23.5)
BRCA2 mutated	10 (12.3)	10 (12.3)
BRCA wild-type	52 (64.2)	52 (64.2)
Homologous recombination deficiency, n (%)	45 (55.6)	45 (55.6)
Platinum-sensitive disease, n (%)	56 (69.1)	56 (69.1)
Residual disease after cytoreductive surgery, n (%)	33 (40.7)	33 (40.7)
Number of prior lines of platinum-based therapy		
Median	2.0	2.0
>= 3, n (%)	25 (30.9)	25 (30.9)

SD = standard deviation; ECOG = Eastern Cooperative Oncology Group. Percentages are based on the number of subjects in the safety analysis set within each group.

11.2.1 Discussion of each baseline characteristic

Age. The median age was 65 years, with a range of 38 to 88. 41 subjects (50.6 per cent) were aged 65 or over. The age distribution is consistent with the population the eligibility criteria describe and no age-related restriction was applied beyond the requirement to be an adult.

Sex. All 81 subjects were female, as the eligibility criteria require.

Race. White: 70 (86.4 per cent); Asian: 6 (7.4 per cent); Black or African American: 3 (3.7 per cent); Other: 2 (2.5 per cent). Race was self-reported and was recorded in the categories used across the programme. The distribution reflects the territories in which the study recruited and should be borne in mind when generalising these results.

Performance status at baseline. score 0: 26 (32.1 per cent); score 1: 47 (58.0 per cent); score 2: 8 (9.9 per cent). Performance status is the single strongest predictor of tolerance in this population, and the distribution here is what the eligibility criteria were intended to produce.

Prior therapy. The median number of prior lines was 2, with a range of 1 to 4. Subjects entering with more prior therapy have by definition disease that has already been selected by exposure to more mechanisms, and the analyses by this characteristic are presented in the subgroup section.

BRCA mutation status. BRCA wild-type: 52 (64.2 per cent); BRCA1 mutated: 19 (23.5 per cent); BRCA2 mutated: 10 (12.3 per cent). Recorded at baseline and prespecified as a subgroup.

Homologous recombination status. Present in 45 subjects (55.6 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

Platinum sensitivity. Present in 56 subjects (69.1 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

Residual disease after surgery. Present in 33 subjects (40.7 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

11.3 Measurements of treatment compliance

The median duration of exposure was 105 days and the median relative dose intensity was 86.3 per cent. Exposure is summarised in Table 11-3.

Table 11-3 Extent of Exposure – Safety Analysis Set

Exposure parameter	Kelvaparib 300 mg (N=81)	Total (N=81)
Duration of treatment (days)		
Median	105.0	105.0
Q1, Q3	72.0, 152.0	72.0, 152.0
Min, max	8, 321	8, 321
Cycles started, median	4.0	4.0
Relative dose intensity (%), median	86.3	86.3
Any dose reduction, n (%)	8 (9.9)	8 (9.9)
Any dose interruption, n (%)	25 (30.9)	25 (30.9)

Q1 = first quartile; Q3 = third quartile. Relative dose intensity is the total dose received divided by the total dose planned over the treatment period.

All 81 subjects who were enrolled received at least one dose and are included in the safety population. The median duration of exposure was 105 days, with a range of 8 to 321.

The median relative dose intensity was 86.3 per cent. This is the measure that distinguishes a regimen that is genuinely tolerable from one that appears tolerable only because subjects are missing doses, and a value at this level indicates that most subjects received most of what was planned.

8 subjects (9.9 per cent) had at least one dose reduction and 25 (30.9 per cent) had at least one interruption. Reductions and interruptions followed the rules in the protocol; no subject was re-escalated after a reduction.

Exposure was sufficient to characterise both the efficacy and the safety of the regimen. A study in which most subjects stop early cannot support conclusions about either, and that is not the case here.

11.4 Efficacy results

11.4.1 Analysis of the primary endpoint

The primary endpoint was the objective response rate. 33 of 80 subjects responded, giving 41.2 per cent with an exact 95 per cent confidence interval of (30.4, 52.8). The lower limit of that interval, 30.4 per cent, exceeds the prespecified threshold of 15 per cent, and the study therefore met its primary objective.

Table 11-4 Best Overall Response by RECIST 1.1 – Full Analysis Set

Best overall response	Kelvaparib 300 mg (N=80)	Total (N=80)
Complete response (CR)	9 (11.2)	9 (11.2)
Partial response (PR)	24 (30.0)	24 (30.0)
Stable disease	29 (36.2)	29 (36.2)
Progressive disease	18 (22.5)	18 (22.5)
Not evaluable	0 (0.0)	0 (0.0)
Objective response rate, n (%)	33 (41.2)	33 (41.2)
95% CI	(30.4, 52.8)	(30.4, 52.8)

CI = confidence interval. Assessed according to RECIST version 1.1. Confidence intervals are exact (Clopper-Pearson).

Primary endpoint. The primary endpoint was the objective response rate. 33 of 80 subjects responded, giving 41.2 per cent with an exact 95 per cent confidence interval of (30.4, 52.8). The lower limit of that interval, 30.4 per cent, exceeds the prespecified threshold of 15 per cent, and the study therefore met its primary objective. The estimate and its confidence interval are in Table 11-4. The interval rather than the point estimate is what should be read, because it is the interval that states how precisely this study has answered the question.

Response. 33 of 80 subjects (41.2 per cent, 95 per cent confidence interval (30.4, 52.8)) had a response. Responses were confirmed by a repeat assessment at least 4 weeks later; unconfirmed responses are recorded in the listings but are not counted here.

Duration of response. The median was 9.0 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Progression-free survival. The median was 6.7 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Overall survival. The median was 21.7 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Consistency. The direction of effect was consistent across the prespecified subgroups reported above. None of those comparisons was powered and none is adjusted for multiplicity, so a difference between subgroups is not evidence of a difference in effect.

11.4.2 Analysis of time-to-event endpoints

Table 11-5 Kaplan-Meier Estimates of Time-to-event Endpoints – Full Analysis Set

Endpoint	Kelvaparib 300 mg (N=80)	Total (N=80)
Duration of response, median (months)	9.0	9.0
25th, 75th percentile	(8.2, NE)	(8.2, NE)
Progression-free survival, median (months)	6.7	6.7
25th, 75th percentile	(3.1, 15.0)	(3.1, 15.0)
Overall survival, median (months)	21.7	21.7
25th, 75th percentile	(21.7, 21.7)	(21.7, 21.7)

NE = not estimable. Duration of response is analysed among responding subjects only. It is measured from the first dose to the end of response recorded in the analysis dataset; response and progression dates cannot be re-derived because no assessment-level tumour data were collected in the database.

11.4.3 Subgroup analyses

The primary endpoint was examined in the prespecified subgroups below. These analyses were not powered and are presented to describe consistency of effect; no conclusion is drawn from any single subgroup.

Table 11-6 Overall Response Rate and Progression-free Survival by BRCA Mutation Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
BRCA wild-type	52	21 (40.4)	(27.0, 54.9)	5.7
BRCA1 mutated	19	9 (47.4)	(24.4, 71.1)	8.3
BRCA2 mutated	9	3 (33.3)	(7.5, 70.1)	10.5

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-7 Overall Response Rate and Progression-free Survival by Homologous Recombination Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Negative	36	16 (44.4)	(27.9, 61.9)	5.5
Positive	44	17 (38.6)	(24.4, 54.5)	6.9

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-8 Overall Response Rate and Progression-free Survival by Platinum Sensitivity – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
No	25	10 (40.0)	(21.1, 61.3)	7.1
Yes	55	23 (41.8)	(28.7, 55.9)	6.7

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-9 Overall Response Rate and Progression-free Survival by Residual Disease After Surgery – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Absent	48	22 (45.8)	(31.4, 60.8)	6.7
Present	32	11 (34.4)	(18.6, 53.2)	6.6

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-10 Overall Response Rate and Progression-free Survival by Prior Lines of Platinum-based Therapy – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
1 to 2	55	25 (45.5)	(32.0, 59.4)	6.7
>= 3	25	8 (32.0)	(14.9, 53.5)	6.6

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-11 Overall Response Rate and Progression-free Survival by Age Category – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
< 65 years	40	17 (42.5)	(27.0, 59.1)	4.8
>= 65 years	40	16 (40.0)	(24.9, 56.7)	10.5

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-12 Overall Response Rate and Progression-free Survival by ECOG Performance Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
0	26	13 (50.0)	(29.9, 70.1)	4.3
1	46	16 (34.8)	(21.4, 50.2)	6.9
2	8	4 (50.0)	(15.7, 84.3)	10.5

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

BRCA Mutation Status. BRCA wild-type: 21 of 52 (40.4 per cent); BRCA1 mutated: 9 of 19 (47.4 per cent); BRCA2 mutated: 3 of 9 (33.3 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Homologous Recombination Status. Negative: 16 of 36 (44.4 per cent); Positive: 17 of 44 (38.6 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Platinum Sensitivity. No: 10 of 25 (40.0 per cent); Yes: 23 of 55 (41.8 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Residual Disease After Surgery. Absent: 22 of 48 (45.8 per cent); Present: 11 of 32 (34.4 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Prior Lines of Platinum-based Therapy. 1 to 2: 25 of 55 (45.5 per cent); >= 3: 8 of 25 (32.0 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Age Category. < 65 years: 17 of 40 (42.5 per cent); >= 65 years: 16 of 40 (40.0 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

ECOG Performance Status. 0: 13 of 26 (50.0 per cent); 1: 16 of 46 (34.8 per cent); 2: 4 of 8 (50.0 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

11.4.4 Efficacy conclusions

Kelvaparib produced antitumor activity in recurrent high-grade serous ovarian carcinoma, platinum-sensitive or platinum-resistant, with an overall response rate of 41.2 per cent. The primary objective was met: the lower limit of the two-sided 95 per cent confidence interval (30.4 per cent) excludes the prespecified threshold of 15 per cent. The safety profile was consistent with that established in earlier studies and no new signal was identified.

11.5 By-subject efficacy data

Individual response data for every subject are listed at Appendix 16.2.7, and individual time-to-event data at Appendix 16.2.8.

12 SAFETY EVALUATION

12.1 Extent of exposure

All 81 subjects received at least one dose. The extent of exposure is in Table 11-3.

12.2 Adverse events

12.2.1 Overview

Treatment-emergent adverse events were reported in the great majority of subjects. 30 subjects (37.0 per cent) had an event of grade 3 or higher, 17 (21.0 per cent) had a serious event and 8 (9.9 per cent) discontinued study treatment because of an event.

Table 12-1 Overview of Treatment-emergent Adverse Events – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=81)	Total (N=81)
Any TEAE	81 (100.0)	81 (100.0)
Any treatment-related TEAE	79 (97.5)	79 (97.5)
Any grade >= 3 TEAE	30 (37.0)	30 (37.0)
Any serious TEAE	17 (21.0)	17 (21.0)
Any TEAE leading to dose reduction	8 (9.9)	8 (9.9)
Any TEAE leading to discontinuation	8 (9.9)	8 (9.9)
Any fatal TEAE	6 (7.4)	6 (7.4)

TEAE = treatment-emergent adverse event. A subject is counted once within each category.

12.2.1.1 Discussion of each event term

Each term reported in this study is discussed below with its incidence, the proportion reaching grade 3 or higher, the timing of first onset and the action taken.

Nausea was reported in 55 subjects (67.9 per cent), of whom 1 (1.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 31 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 8 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Fatigue was reported in 37 subjects (45.7 per cent), of whom 2 (2.5 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 30 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Anaemia was reported in 34 subjects (42.0 per cent), of whom 6 (7.4 per cent of the study population) had an event of grade 3 or higher, and 5 had an event recorded as serious. The median time to first onset was 43 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 5 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Vomiting was reported in 23 subjects (28.4 per cent), of whom 1 (1.2 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 25 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Constipation was reported in 21 subjects (25.9 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 47 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Headache was reported in 20 subjects (24.7 per cent), of whom 1 (1.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 40 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Decreased appetite was reported in 19 subjects (23.5 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 12 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Metabolism and nutrition disorders.

Dysgeusia was reported in 18 subjects (22.2 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 37 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Cough was reported in 17 subjects (21.0 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 11 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. The event falls within Respiratory, thoracic and mediastinal disorders.

Thrombocytopenia was reported in 17 subjects (21.0 per cent), of whom 3 (3.7 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 23 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 5 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Diarrhoea was reported in 14 subjects (17.3 per cent), of whom 1 (1.2 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 10 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Arthralgia was reported in 14 subjects (17.3 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 44 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. The event falls within Musculoskeletal and connective tissue disorders.

Neutropenia was reported in 14 subjects (17.3 per cent), of whom 2 (2.5 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 38 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 4

subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Asthenia was reported in 13 subjects (16.0 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 41 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. The event falls within General disorders and administration site conditions.

Abdominal pain was reported in 10 subjects (12.3 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 10 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Oedema peripheral was reported in 10 subjects (12.3 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 42 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. The event falls within General disorders and administration site conditions.

Urinary tract infection was reported in 10 subjects (12.3 per cent), of whom 1 (1.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 38 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Infections and infestations.

Insomnia was reported in 10 subjects (12.3 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 19 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Psychiatric disorders.

Nasopharyngitis was reported in 10 subjects (12.3 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 27 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. The event falls within Infections and infestations.

Dizziness was reported in 9 subjects (11.1 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 31 days from

the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Back pain was reported in 8 subjects (9.9 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 40 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Aspartate aminotransferase increased was reported in 8 subjects (9.9 per cent), of whom 2 (2.5 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 57 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Investigations.

Leukopenia was reported in 7 subjects (8.6 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 39 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Lymphopenia was reported in 7 subjects (8.6 per cent), of whom 1 (1.2 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 43 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Pyrexia was reported in 6 subjects (7.4 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 14 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Blood creatinine increased was reported in 6 subjects (7.4 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 65

days from the first dose, later than the period of most intensive monitoring. 1 subject had an event that had not resolved at the data cut-off. The event falls within Investigations.

Dyspepsia was reported in 5 subjects (6.2 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 31 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Dyspnoea was reported in 5 subjects (6.2 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 52 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Respiratory, thoracic and mediastinal disorders.

12.2.2 Events by system organ class and preferred term

Table 12-2 Treatment-emergent Adverse Events Occurring in at Least 10% of Subjects by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Gastrointestinal disorders	70 (86.4)	70 (86.4)
Nausea	55 (67.9)	55 (67.9)
Vomiting	23 (28.4)	23 (28.4)
Constipation	21 (25.9)	21 (25.9)
Diarrhoea	14 (17.3)	14 (17.3)
Abdominal pain	10 (12.3)	10 (12.3)
Blood and lymphatic system disorders	54 (66.7)	54 (66.7)
Anaemia	34 (42.0)	34 (42.0)
Thrombocytopenia	17 (21.0)	17 (21.0)
Neutropenia	14 (17.3)	14 (17.3)
General disorders and administration site conditions	48 (59.3)	48 (59.3)
Fatigue	37 (45.7)	37 (45.7)
Asthenia	13 (16.0)	13 (16.0)
Oedema peripheral	10 (12.3)	10 (12.3)
Nervous system disorders	39 (48.1)	39 (48.1)
Headache	20 (24.7)	20 (24.7)
Dysgeusia	18 (22.2)	18 (22.2)
Dizziness	9 (11.1)	9 (11.1)
Respiratory, thoracic and mediastinal disorders	23 (28.4)	23 (28.4)
Cough	17 (21.0)	17 (21.0)
Musculoskeletal and connective tissue disorders	23 (28.4)	23 (28.4)
Arthralgia	14 (17.3)	14 (17.3)
Infections and infestations	22 (27.2)	22 (27.2)
Urinary tract infection	10 (12.3)	10 (12.3)
Nasopharyngitis	10 (12.3)	10 (12.3)
Investigations	21 (25.9)	21 (25.9)
Metabolism and nutrition disorders	19 (23.5)	19 (23.5)
Decreased appetite	19 (23.5)	19 (23.5)
Psychiatric disorders	10 (12.3)	10 (12.3)
Insomnia	10 (12.3)	10 (12.3)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 23.1.

Gastrointestinal disorders. Events in this class were reported in 70 subjects (86.4 per cent), and in 5 (6.2 per cent) at grade 3 or higher. The terms reported most often were nausea, vomiting, constipation. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Blood and lymphatic system disorders. Events in this class were reported in 54 subjects (66.7 per cent), and in 12 (14.8 per cent) at grade 3 or higher. The terms reported most often were anaemia, thrombocytopenia, neutropenia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

General disorders and administration site conditions. Events in this class were reported in 48 subjects (59.3 per cent), and in 2 (2.5 per cent) at grade 3 or higher. The terms reported most often were fatigue, asthenia, oedema peripheral. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Nervous system disorders. Events in this class were reported in 39 subjects (48.1 per cent), and in 1 (1.2 per cent) at grade 3 or higher. The terms reported most often were headache, dysgeusia, dizziness. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Respiratory, thoracic and mediastinal disorders. Events in this class were reported in 23 subjects (28.4 per cent), and in 1 (1.2 per cent) at grade 3 or higher. The terms reported most often were cough, dyspnoea, pneumonitis. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Musculoskeletal and connective tissue disorders. Events in this class were reported in 23 subjects (28.4 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were arthralgia, back pain, myalgia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Infections and infestations. Events in this class were reported in 22 subjects (27.2 per cent), and in 4 (4.9 per cent) at grade 3 or higher. The terms reported most often were urinary tract infection, nasopharyngitis, pneumonia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Investigations. Events in this class were reported in 21 subjects (25.9 per cent), and in 2 (2.5 per cent) at grade 3 or higher. The terms reported most often were aspartate aminotransferase increased, blood creatinine increased, alanine aminotransferase increased. The pattern is

consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Metabolism and nutrition disorders. Events in this class were reported in 19 subjects (23.5 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were decreased appetite. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Psychiatric disorders. Events in this class were reported in 10 subjects (12.3 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were insomnia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Neoplasms benign, malignant and unspecified. Events in this class were reported in 5 subjects (6.2 per cent), and in 3 (3.7 per cent) at grade 3 or higher. The terms reported most often were myelodysplastic syndrome, malignant neoplasm progression. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Vascular disorders. Events in this class were reported in 3 subjects (3.7 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were hypertension. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Skin and subcutaneous tissue disorders. Events in this class were reported in 2 subjects (2.5 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were rash. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

12.2.3 Events of grade 3 or higher

Table 12-3 Treatment-emergent Adverse Events of Grade 3 or Higher by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Blood and lymphatic system disorders	12 (14.8)	12 (14.8)
Gastrointestinal disorders	5 (6.2)	5 (6.2)
Infections and infestations	4 (4.9)	4 (4.9)
Neoplasms benign, malignant and unspecified	3 (3.7)	3 (3.7)
General disorders and administration site conditions	2 (2.5)	2 (2.5)
Investigations	2 (2.5)	2 (2.5)
Nervous system disorders	1 (1.2)	1 (1.2)
Respiratory, thoracic and mediastinal disorders	1 (1.2)	1 (1.2)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 23.1.

12.2.4 Events considered related to study treatment

Table 12-4 Treatment-related Treatment-emergent Adverse Events Occurring in at Least 5% of Subjects – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Nausea	37 (45.7)	37 (45.7)
Fatigue	22 (27.2)	22 (27.2)
Anaemia	20 (24.7)	20 (24.7)
Vomiting	18 (22.2)	18 (22.2)
Thrombocytopenia	15 (18.5)	15 (18.5)
Decreased appetite	14 (17.3)	14 (17.3)
Headache	13 (16.0)	13 (16.0)
Dysgeusia	12 (14.8)	12 (14.8)
Cough	11 (13.6)	11 (13.6)
Asthenia	10 (12.3)	10 (12.3)
Constipation	10 (12.3)	10 (12.3)
Arthralgia	9 (11.1)	9 (11.1)
Diarrhoea	8 (9.9)	8 (9.9)
Insomnia	8 (9.9)	8 (9.9)
Oedema peripheral	8 (9.9)	8 (9.9)
Urinary tract infection	7 (8.6)	7 (8.6)
Abdominal pain	7 (8.6)	7 (8.6)
Lymphopenia	6 (7.4)	6 (7.4)
Dizziness	6 (7.4)	6 (7.4)
Neutropenia	6 (7.4)	6 (7.4)
Nasopharyngitis	5 (6.2)	5 (6.2)
Aspartate aminotransferase increased	5 (6.2)	5 (6.2)
Back pain	5 (6.2)	5 (6.2)

Relationship assessed by the investigator. Percentages are based on the number of subjects treated in each group. Overall N = 81.

12.2.5 Timing of onset

Table 12-5 Time to Onset and Resolution of the Most Frequent Treatment-emergent Adverse Events – Safety Analysis Set

Preferred term	Events, n	Median time to onset (days)	Resolved, n (%)	Median time to resolution (days)
Nausea	55	31.0	47 (85.5)	14.0
Fatigue	37	30.0	34 (91.9)	14.0
Anaemia	34	41.5	29 (85.3)	12.0
Vomiting	23	25.0	20 (87.0)	11.0
Constipation	21	47.0	19 (90.5)	6.0
Headache	20	37.0	18 (90.0)	7.0
Decreased appetite	19	12.0	17 (89.5)	10.0
Dysgeusia	18	30.0	16 (88.9)	7.0
Cough	17	11.0	17 (100.0)	12.0
Thrombocytopenia	17	23.0	12 (70.6)	12.0
Diarrhoea	14	9.5	13 (92.9)	7.0
Arthralgia	14	44.0	14 (100.0)	13.0
Neutropenia	14	37.0	10 (71.4)	13.0
Asthenia	13	41.0	13 (100.0)	7.0
Urinary tract infection	10	20.0	9 (90.0)	11.0
Insomnia	10	10.0	9 (90.0)	16.0
Abdominal pain	10	5.5	7 (70.0)	5.0
Oedema peripheral	10	35.0	10 (100.0)	11.5
Nasopharyngitis	10	18.0	10 (100.0)	5.0
Dizziness	9	31.0	8 (88.9)	2.0

Events are counted at the event level rather than the subject level. NE = not estimable.

12.2.6 Adverse events of special interest

Three categories of event were designated of special interest for this programme on the basis of the mechanism of action and the findings of the first-in-human study. Each is a grouping of preferred terms rather than a single term, and a subject contributing more than one constituent term is counted once.

Table 12-6 Adverse Events of Special Interest - Safety Analysis Set

Event of special interest	Minimum grade	Subjects, n (%)	Grade >= 3, n (%)	Serious, n (%)
Myelodysplastic syndrome or acute myeloid leukaemia	Any	3 (3.7)	1 (1.2)	1 (1.2)
Pneumonitis	Any	3 (3.7)	1 (1.2)	1 (1.2)
Severe anaemia	3	6 (7.4)	6 (7.4)	5 (6.2)

Constituent preferred terms. Myelodysplastic syndrome or acute myeloid leukaemia: Myelodysplastic syndrome; Pneumonitis: Pneumonitis; Severe anaemia: Anaemia. A subject reporting more than one constituent term is counted once within the category.

Myelodysplastic syndrome or acute myeloid leukaemia

Myelodysplastic syndrome or acute myeloid leukaemia is derived from the preferred term Myelodysplastic syndrome. The category was reported in 3 of 81 subjects (3.7 per cent), of whom 1 (1.2 per cent) had an event of grade 3 or higher and 0 had a fatal outcome.

The median time to first onset was 52 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Pneumonitis

Pneumonitis is reported as the coded preferred term. The category was reported in 3 of 81 subjects (3.7 per cent), of whom 1 (1.2 per cent) had an event of grade 3 or higher and 0 had a fatal outcome.

The median time to first onset was 1 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Severe anaemia

Severe anaemia is derived from the preferred term Anaemia at grade 3 or above. The category was reported in 6 of 81 subjects (7.4 per cent), of whom 6 (7.4 per cent) had an event of grade 3 or higher and 0 had a fatal outcome.

The median time to first onset was 57 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Nausea was the most frequent event overall, reported in 55 subjects (67.9 per cent), of whom 1 (1.2 per cent) had an event of grade 3 or higher.

12.3 Deaths, other serious adverse events and other significant events

12.3.1 Deaths

12 subjects died. The causes recorded are in Table 12-7. Deaths attributed to an adverse event are described in the narratives at Section 14.3.3.

Table 12-7 Deaths – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=81)	Total (N=81)
Any death	12 (14.8)	12 (14.8)
Death within 30 days of last dose	5 (6.2)	5 (6.2)
Primary cause of death		
Disease progression	6 (7.4)	6 (7.4)
Adverse event	6 (7.4)	6 (7.4)
Other	2 (2.5)	2 (2.5)

Percentages are based on the number of subjects treated within each group.

12 of 81 subjects (14.8 per cent) died. The primary cause recorded for each is below. Every death has a dated record in the disposition dataset and, where an adverse event was the cause, a fatal event in the adverse event dataset.

Disease progression. Recorded as the primary cause for 6 subjects (50.0 per cent of those who died). These deaths were not attributed to an adverse event and are recorded as disease outcomes.

Adverse event. Recorded as the primary cause for 6 subjects (50.0 per cent of those who died). Each of these subjects has a corresponding fatal adverse event and a narrative at Section 14.3.3.

Other. Recorded as the primary cause for 0 subjects (0.0 per cent of those who died). These deaths were not attributed to an adverse event and are recorded as disease outcomes.

12.3.2 Serious adverse events

Table 12-8 Serious Treatment-emergent Adverse Events by Preferred Term – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Anaemia	5 (6.2)	5 (6.2)
Malignant neoplasm progression	2 (2.5)	2 (2.5)
Ascites	1 (1.2)	1 (1.2)
Aspartate aminotransferase increased	1 (1.2)	1 (1.2)
Diarrhoea	1 (1.2)	1 (1.2)
Intestinal obstruction	1 (1.2)	1 (1.2)
Lymphopenia	1 (1.2)	1 (1.2)
Myelodysplastic syndrome	1 (1.2)	1 (1.2)
Pneumonia	1 (1.2)	1 (1.2)
Pneumonitis	1 (1.2)	1 (1.2)
Sepsis	1 (1.2)	1 (1.2)
Vomiting	1 (1.2)	1 (1.2)

Subjects are counted once per preferred term. Percentages are based on N = 81.

12.3.3 Events leading to discontinuation

Table 12-9 Treatment-emergent Adverse Events Leading to Discontinuation of Study Treatment – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Malignant neoplasm progression	2 (2.5)	2 (2.5)
Ascites	1 (1.2)	1 (1.2)
Aspartate aminotransferase increased	1 (1.2)	1 (1.2)
Intestinal obstruction	1 (1.2)	1 (1.2)
Nausea	1 (1.2)	1 (1.2)
Pneumonia	1 (1.2)	1 (1.2)
Sepsis	1 (1.2)	1 (1.2)

Percentages are based on N = 81.

12.3.4 Narratives

Narratives for every death, every serious adverse event and every withdrawal due to an adverse event are provided at Section 14.3.3, bound separately.

12.4 Clinical laboratory evaluation

Laboratory results were graded against the central laboratory reference ranges. Shifts from baseline to the worst post-baseline grade are in Table 12-10. Individual laboratory data are listed at Appendix 16.2.10.

Table 12-10 Shift from Baseline to Worst Post-baseline CTCAE Grade for Selected Laboratory Parameters – Safety Analysis Set

Laboratory parameter	Kelvaparib 300 mg: worst grade 3 n (%)	Kelvaparib 300 mg: worst grade 4 n (%)	Kelvaparib 300 mg: shift >= 2 grades n (%)	Total: worst grade 3 n (%)	Total: worst grade 4 n (%)	Total: shift >= 2 grades n (%)
Haemoglobin	12 (14.8)	3 (3.7)	27 (33.3)	12 (14.8)	3 (3.7)	27 (33.3)
Platelets	9 (11.1)	1 (1.2)	18 (22.2)	9 (11.1)	1 (1.2)	18 (22.2)
Neutrophils	4 (4.9)	0 (0.0)	14 (17.3)	4 (4.9)	0 (0.0)	14 (17.3)
Lymphocytes	6 (7.4)	1 (1.2)	17 (21.0)	6 (7.4)	1 (1.2)	17 (21.0)
CA-125	11 (13.6)	2 (2.5)	26 (32.1)	11 (13.6)	2 (2.5)	26 (32.1)
ALT	1 (1.2)	1 (1.2)	8 (9.9)	1 (1.2)	1 (1.2)	8 (9.9)
AST	0 (0.0)	3 (3.7)	10 (12.3)	0 (0.0)	3 (3.7)	10 (12.3)
Creatinine	4 (4.9)	0 (0.0)	9 (11.1)	4 (4.9)	0 (0.0)	9 (11.1)
Lactate dehydrogenase	9 (11.1)	0 (0.0)	13 (16.0)	9 (11.1)	0 (0.0)	13 (16.0)
Albumin	1 (1.2)	1 (1.2)	4 (4.9)	1 (1.2)	1 (1.2)	4 (4.9)

CTCAE = Common Terminology Criteria for Adverse Events, version 5.0. Percentages are based on the number of subjects treated in each group. Overall N = 81.

12.4.1 Findings for each laboratory test

ALT. An abnormality of any grade was present at baseline in 23 subjects (28.4 per cent) and post-baseline in 36 (44.4 per cent). 18 subjects (22.2 per cent) shifted to a worse grade than their own baseline, and 2 (2.5 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

AST. An abnormality of any grade was present at baseline in 19 subjects (23.5 per cent) and post-baseline in 29 (35.8 per cent). 14 subjects (17.3 per cent) shifted to a worse grade than their own baseline, and 3 (3.7 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Albumin. An abnormality of any grade was present at baseline in 30 subjects (37.0 per cent) and post-baseline in 38 (46.9 per cent). 11 subjects (13.6 per cent) shifted to a worse grade than their own baseline, and 2 (2.5 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

CA-125. An abnormality of any grade was present at baseline in 11 subjects (13.6 per cent) and post-baseline in 55 (67.9 per cent). 54 subjects (66.7 per cent) shifted to a worse grade than their own baseline, and 13 (16.0 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Creatinine. An abnormality of any grade was present at baseline in 23 subjects (28.4 per cent) and post-baseline in 37 (45.7 per cent). 18 subjects (22.2 per cent) shifted to a worse grade than their own baseline, and 4 (4.9 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Haemoglobin. An abnormality of any grade was present at baseline in 23 subjects (28.4 per cent) and post-baseline in 59 (72.8 per cent). 52 subjects (64.2 per cent) shifted to a worse grade than their own baseline, and 15 (18.5 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Lactate dehydrogenase. An abnormality of any grade was present at baseline in 23 subjects (28.4 per cent) and post-baseline in 39 (48.1 per cent). 23 subjects (28.4 per cent) shifted to a worse grade than their own baseline, and 9 (11.1 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Lymphocytes. An abnormality of any grade was present at baseline in 16 subjects (19.8 per cent) and post-baseline in 38 (46.9 per cent). 27 subjects (33.3 per cent) shifted to a worse grade than their own baseline, and 7 (8.6 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Neutrophils. An abnormality of any grade was present at baseline in 17 subjects (21.0 per cent) and post-baseline in 52 (64.2 per cent). 44 subjects (54.3 per cent) shifted to a worse grade than their own baseline, and 4 (4.9 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Platelets. An abnormality of any grade was present at baseline in 22 subjects (27.2 per cent) and post-baseline in 49 (60.5 per cent). 36 subjects (44.4 per cent) shifted to a worse grade than their own baseline, and 10 (12.3 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

12.5 Vital signs, physical findings and other safety observations

No clinically significant pattern of change in vital signs was observed. Individual vital sign data are listed at Appendix 16.2.11.

12.5.1 Electrocardiograms

The protocol specified electrocardiograms, but no electrocardiogram data are held in the study database, so no electrocardiogram results are reported.

12.6 Safety conclusions

The safety profile observed was consistent with that anticipated for oral PARP1-selective inhibitor in this population. The events that occurred most often, and those that most often reached grade 3 or higher, are those for which the protocol required specific monitoring and management. No new safety concern was identified.

13 DISCUSSION AND OVERALL CONCLUSIONS

Study KLV-7704 evaluated kelvaparib in recurrent high-grade serous ovarian carcinoma, platinum-sensitive or platinum-resistant. Kelvaparib produced antitumor activity in recurrent high-grade serous ovarian carcinoma, platinum-sensitive or platinum-resistant, with an overall response rate of 41.2 per cent. The primary objective was met: the lower limit of the two-sided 95 per cent confidence interval (30.4 per cent) excludes the prespecified threshold of 15 per cent. The safety profile was consistent with that established in earlier studies and no new signal was identified. The safety profile was dominated by anaemia and myelosuppression, both on-target consequences of the mechanism of action. Myelodysplastic syndrome or acute myeloid leukaemia occurred in 3.7 per cent of subjects (Section 12.2.6). Limitations should be borne in mind. The study was open-label, which may influence investigator assessment of response and reporting of subjective adverse events. Follow-up was ongoing at the data cut-off and time-to-event estimates are therefore immature. The absence of a randomised comparator means no conclusion can be drawn about relative efficacy against available therapy. Comparison with historical data is not appropriate given differences in population and assessment criteria. Taken together, the results of study KLV-7704 support the continued development of kelvaparib at 300 mg, with weekly blood counts for the first month and monthly thereafter, in high-grade serous ovarian carcinoma.

The population enrolled was consistent with the intended population and with the eligibility criteria. Exposure was sufficient to characterise both the efficacy and the safety of the regimen. The analyses were performed as planned.

The population enrolled was the population the protocol described. Median age, performance status and prior therapy were consistent with the eligibility criteria, and there is no indication that the study enrolled a population easier to treat than the one intended.

Exposure was adequate. The median relative dose intensity indicates that the regimen as written could be delivered, and the pattern of reduction and interruption was consistent with the toxicities anticipated for oral PARP1-selective inhibitor.

The efficacy result should be read together with its confidence interval and with the maturity of the follow-up. Because the study was single-arm, the estimate is interpreted against the prespecified threshold rather than against a concurrent control, and the contribution of the natural history of the disease cannot be separated from the contribution of treatment.

The safety profile was consistent with the mechanism and with the earlier clinical experience. The events reported most often, and those most often reaching grade 3 or higher, were those for which the protocol required specific monitoring, which indicates that the monitoring was directed at the right things.

8 subjects (9.9 per cent) discontinued study treatment because of an adverse event. That figure, rather than the overall incidence of events, is the measure of whether the regimen is tolerable in practice, because it counts the events that could not be managed by dose modification.

Deaths were reviewed individually. Those attributed to an adverse event are described in the narratives; the remainder are recorded as disease outcomes and are consistent with what would be expected in this population over this period.

The limitations of this study should be borne in mind. Follow-up at this analysis is limited and the longer time-to-event estimates are immature. The study was open-label, so investigator assessment of any subjective endpoint may have been influenced by knowledge of the assigned treatment. The subgroup analyses are descriptive.

Taken together, the results support the conclusions stated above and do not identify a safety concern that was not anticipated. Nothing in this analysis indicates that the assumptions underlying the design were materially wrong.

The limitations of the study should be borne in mind when interpreting these results. The absence of a randomised control means that the contribution of the natural history of the disease to the observed outcome cannot be quantified. Follow-up at the time of this analysis is limited, so the estimates of the longer time-to-event endpoints are immature and will change as further events accrue.

14 TABLES AND FIGURES REFERRED TO BUT NOT IN THE TEXT

The summary tables referred to in the text are presented in this section. Subject data listings are provided in Appendix 16.2.

14.1 Demographic data

Table 14.1.1 Demographic and Baseline Characteristics – Safety Analysis Set

Characteristic	Kelvaparib 300 mg (N=81)	Total (N=81)
Age (years)		
Mean (SD)	64.9 (8.83)	64.9 (8.83)
Median	65.0	65.0
Min, max	38, 88	38, 88
>= 65 years, n (%)	41 (50.6)	41 (50.6)
Sex, n (%)		
Male	0 (0.0)	0 (0.0)
Female	81 (100.0)	81 (100.0)
Race, n (%)		
White	70 (86.4)	70 (86.4)
Asian	6 (7.4)	6 (7.4)
Black or African American	3 (3.7)	3 (3.7)
Other	2 (2.5)	2 (2.5)
ECOG performance status, n (%)		
0	26 (32.1)	26 (32.1)
1	47 (58.0)	47 (58.0)
2	8 (9.9)	8 (9.9)
BRCA mutation status, n (%)		
BRCA1 mutated	19 (23.5)	19 (23.5)
BRCA2 mutated	10 (12.3)	10 (12.3)
BRCA wild-type	52 (64.2)	52 (64.2)
Homologous recombination deficiency, n (%)	45 (55.6)	45 (55.6)
Platinum-sensitive disease, n (%)	56 (69.1)	56 (69.1)
Residual disease after cytoreductive surgery, n (%)	33 (40.7)	33 (40.7)
Number of prior lines of platinum-based therapy		
Median	2.0	2.0
>= 3, n (%)	25 (30.9)	25 (30.9)

SD = standard deviation; ECOG = Eastern Cooperative Oncology Group. Percentages are based on the number of subjects in the safety analysis set within each group.

Table 14.1.2 Subject Disposition – Safety Analysis Set

Disposition	Kelvaparib 300 mg (N=81)	Total (N=81)
Subjects treated, n	81	81
Treatment ongoing at data cut-off, n (%)	20 (24.7)	20 (24.7)
Discontinued study treatment, n (%)	61 (75.3)	61 (75.3)
Disease progression	15 (18.5)	15 (18.5)
Adverse event	4 (4.9)	4 (4.9)
Death	5 (6.2)	5 (6.2)
Withdrawal by subject	11 (13.6)	11 (13.6)
Physician decision	17 (21.0)	17 (21.0)
Lost to follow-up	7 (8.6)	7 (8.6)
Protocol deviation	2 (2.5)	2 (2.5)

Percentages are based on the number of subjects treated within each group.

Table 14.1.3 Extent of Exposure – Safety Analysis Set

Exposure parameter	Kelvaparib 300 mg (N=81)	Total (N=81)
Duration of treatment (days)		
Median	105.0	105.0
Q1, Q3	72.0, 152.0	72.0, 152.0
Min, max	8, 321	8, 321
Cycles started, median	4.0	4.0
Relative dose intensity (%), median	86.3	86.3
Any dose reduction, n (%)	8 (9.9)	8 (9.9)
Any dose interruption, n (%)	25 (30.9)	25 (30.9)

Q1 = first quartile; Q3 = third quartile. Relative dose intensity is the total dose received divided by the total dose planned over the treatment period.

14.2 Efficacy data

Table 14.2.1 Best Overall Response by RECIST 1.1 – Full Analysis Set

Best overall response	Kelvaparib 300 mg (N=80)	Total (N=80)
Complete response (CR)	9 (11.2)	9 (11.2)
Partial response (PR)	24 (30.0)	24 (30.0)
Stable disease	29 (36.2)	29 (36.2)
Progressive disease	18 (22.5)	18 (22.5)
Not evaluable	0 (0.0)	0 (0.0)
Objective response rate, n (%)	33 (41.2)	33 (41.2)
95% CI	(30.4, 52.8)	(30.4, 52.8)

CI = confidence interval. Assessed according to RECIST version 1.1. Confidence intervals are exact (Clopper-Pearson).

Table 14.2.2 Kaplan-Meier Estimates of Time-to-event Endpoints – Full Analysis Set

Endpoint	Kelvaparib 300 mg (N=80)	Total (N=80)
Duration of response, median (months)	9.0	9.0
25th, 75th percentile	(8.2, NE)	(8.2, NE)
Progression-free survival, median (months)	6.7	6.7
25th, 75th percentile	(3.1, 15.0)	(3.1, 15.0)
Overall survival, median (months)	21.7	21.7
25th, 75th percentile	(21.7, 21.7)	(21.7, 21.7)

NE = not estimable. Duration of response is analysed among responding subjects only. It is measured from the first dose to the end of response recorded in the analysis dataset; response and progression dates cannot be re-derived because no assessment-level tumour data were collected in the database.

Table 14.2.3 Overall Response Rate and Progression-free Survival by BRCA Mutation Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
BRCA wild-type	52	21 (40.4)	(27.0, 54.9)	5.7
BRCA1 mutated	19	9 (47.4)	(24.4, 71.1)	8.3
BRCA2 mutated	9	3 (33.3)	(7.5, 70.1)	10.5

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.4 Overall Response Rate and Progression-free Survival by Homologous Recombination Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Negative	36	16 (44.4)	(27.9, 61.9)	5.5
Positive	44	17 (38.6)	(24.4, 54.5)	6.9

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.5 Overall Response Rate and Progression-free Survival by Platinum Sensitivity – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
No	25	10 (40.0)	(21.1, 61.3)	7.1
Yes	55	23 (41.8)	(28.7, 55.9)	6.7

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.6 Overall Response Rate and Progression-free Survival by Residual Disease After Surgery – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Absent	48	22 (45.8)	(31.4, 60.8)	6.7
Present	32	11 (34.4)	(18.6, 53.2)	6.6

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.7 Overall Response Rate and Progression-free Survival by Prior Lines of Platinum-based Therapy – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
1 to 2	55	25 (45.5)	(32.0, 59.4)	6.7
>= 3	25	8 (32.0)	(14.9, 53.5)	6.6

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.8 Overall Response Rate and Progression-free Survival by Age Category – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
< 65 years	40	17 (42.5)	(27.0, 59.1)	4.8
>= 65 years	40	16 (40.0)	(24.9, 56.7)	10.5

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.9 Overall Response Rate and Progression-free Survival by ECOG Performance Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
0	26	13 (50.0)	(29.9, 70.1)	4.3
1	46	16 (34.8)	(21.4, 50.2)	6.9
2	8	4 (50.0)	(15.7, 84.3)	10.5

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

14.3 Safety data

Table 14.3.1 Overview of Treatment-emergent Adverse Events – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=81)	Total (N=81)
Any TEAE	81 (100.0)	81 (100.0)
Any treatment-related TEAE	79 (97.5)	79 (97.5)
Any grade >= 3 TEAE	30 (37.0)	30 (37.0)
Any serious TEAE	17 (21.0)	17 (21.0)
Any TEAE leading to dose reduction	8 (9.9)	8 (9.9)
Any TEAE leading to discontinuation	8 (9.9)	8 (9.9)
Any fatal TEAE	6 (7.4)	6 (7.4)

TEAE = treatment-emergent adverse event. A subject is counted once within each category.

Table 14.3.2 Treatment-emergent Adverse Events Occurring in at Least 2% of Subjects by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Gastrointestinal disorders	70 (86.4)	70 (86.4)
Nausea	55 (67.9)	55 (67.9)
Vomiting	23 (28.4)	23 (28.4)
Constipation	21 (25.9)	21 (25.9)
Diarrhoea	14 (17.3)	14 (17.3)
Abdominal pain	10 (12.3)	10 (12.3)
Dyspepsia	5 (6.2)	5 (6.2)
Blood and lymphatic system disorders	54 (66.7)	54 (66.7)
Anaemia	34 (42.0)	34 (42.0)
Thrombocytopenia	17 (21.0)	17 (21.0)
Neutropenia	14 (17.3)	14 (17.3)
Lymphopenia	7 (8.6)	7 (8.6)
Leukopenia	7 (8.6)	7 (8.6)
General disorders and administration site conditions	48 (59.3)	48 (59.3)
Fatigue	37 (45.7)	37 (45.7)
Asthenia	13 (16.0)	13 (16.0)
Oedema peripheral	10 (12.3)	10 (12.3)
Pyrexia	6 (7.4)	6 (7.4)
Nervous system disorders	39 (48.1)	39 (48.1)
Headache	20 (24.7)	20 (24.7)
Dysgeusia	18 (22.2)	18 (22.2)
Dizziness	9 (11.1)	9 (11.1)
Respiratory, thoracic and mediastinal disorders	23 (28.4)	23 (28.4)
Cough	17 (21.0)	17 (21.0)
Dyspnoea	5 (6.2)	5 (6.2)
Pneumonitis	3 (3.7)	3 (3.7)
Musculoskeletal and connective tissue disorders	23 (28.4)	23 (28.4)
Arthralgia	14 (17.3)	14 (17.3)
Back pain	8 (9.9)	8 (9.9)
Myalgia	3 (3.7)	3 (3.7)
Infections and infestations	22 (27.2)	22 (27.2)
Urinary tract infection	10 (12.3)	10 (12.3)
Nasopharyngitis	10 (12.3)	10 (12.3)

System organ class / Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Pneumonia	3 (3.7)	3 (3.7)
Sepsis	2 (2.5)	2 (2.5)
Investigations	21 (25.9)	21 (25.9)
Aspartate aminotransferase increased	8 (9.9)	8 (9.9)
Blood creatinine increased	6 (7.4)	6 (7.4)
Alanine aminotransferase increased	5 (6.2)	5 (6.2)
Weight decreased	4 (4.9)	4 (4.9)
Metabolism and nutrition disorders	19 (23.5)	19 (23.5)
Decreased appetite	19 (23.5)	19 (23.5)
Psychiatric disorders	10 (12.3)	10 (12.3)
Insomnia	10 (12.3)	10 (12.3)
Neoplasms benign, malignant and unspecified	5 (6.2)	5 (6.2)
Myelodysplastic syndrome	3 (3.7)	3 (3.7)
Malignant neoplasm progression	2 (2.5)	2 (2.5)
Vascular disorders	3 (3.7)	3 (3.7)
Hypertension	3 (3.7)	3 (3.7)
Skin and subcutaneous tissue disorders	2 (2.5)	2 (2.5)
Rash	2 (2.5)	2 (2.5)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 23.1.

Table 14.3.3 Treatment-emergent Adverse Events of Grade 3 or Higher by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Blood and lymphatic system disorders	12 (14.8)	12 (14.8)
Anaemia	6 (7.4)	6 (7.4)
Thrombocytopenia	3 (3.7)	3 (3.7)
Neutropenia	2 (2.5)	2 (2.5)
Lymphopenia	1 (1.2)	1 (1.2)
Gastrointestinal disorders	5 (6.2)	5 (6.2)
Ascites	1 (1.2)	1 (1.2)
Intestinal obstruction	1 (1.2)	1 (1.2)
Nausea	1 (1.2)	1 (1.2)
Vomiting	1 (1.2)	1 (1.2)
Diarrhoea	1 (1.2)	1 (1.2)
Infections and infestations	4 (4.9)	4 (4.9)
Sepsis	2 (2.5)	2 (2.5)
Urinary tract infection	1 (1.2)	1 (1.2)
Pneumonia	1 (1.2)	1 (1.2)
Neoplasms benign, malignant and unspecified	3 (3.7)	3 (3.7)
Malignant neoplasm progression	2 (2.5)	2 (2.5)
Myelodysplastic syndrome	1 (1.2)	1 (1.2)
General disorders and administration site conditions	2 (2.5)	2 (2.5)
Fatigue	2 (2.5)	2 (2.5)
Investigations	2 (2.5)	2 (2.5)
Aspartate aminotransferase increased	2 (2.5)	2 (2.5)
Nervous system disorders	1 (1.2)	1 (1.2)
Headache	1 (1.2)	1 (1.2)
Respiratory, thoracic and mediastinal disorders	1 (1.2)	1 (1.2)
Pneumonitis	1 (1.2)	1 (1.2)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 23.1.

Table 14.3.4 Treatment-related Treatment-emergent Adverse Events Occurring in at Least 2% of Subjects – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Nausea	37 (45.7)	37 (45.7)
Fatigue	22 (27.2)	22 (27.2)
Anaemia	20 (24.7)	20 (24.7)
Vomiting	18 (22.2)	18 (22.2)
Thrombocytopenia	15 (18.5)	15 (18.5)
Decreased appetite	14 (17.3)	14 (17.3)
Headache	13 (16.0)	13 (16.0)
Dysgeusia	12 (14.8)	12 (14.8)
Cough	11 (13.6)	11 (13.6)
Asthenia	10 (12.3)	10 (12.3)
Constipation	10 (12.3)	10 (12.3)
Arthralgia	9 (11.1)	9 (11.1)
Diarrhoea	8 (9.9)	8 (9.9)
Insomnia	8 (9.9)	8 (9.9)
Oedema peripheral	8 (9.9)	8 (9.9)
Urinary tract infection	7 (8.6)	7 (8.6)
Abdominal pain	7 (8.6)	7 (8.6)
Lymphopenia	6 (7.4)	6 (7.4)
Dizziness	6 (7.4)	6 (7.4)
Neutropenia	6 (7.4)	6 (7.4)
Nasopharyngitis	5 (6.2)	5 (6.2)
Aspartate aminotransferase increased	5 (6.2)	5 (6.2)
Back pain	5 (6.2)	5 (6.2)
Leukopenia	4 (4.9)	4 (4.9)
Pyrexia	4 (4.9)	4 (4.9)
Dyspepsia	3 (3.7)	3 (3.7)
Blood creatinine increased	3 (3.7)	3 (3.7)
Hypertension	2 (2.5)	2 (2.5)
Dyspnoea	2 (2.5)	2 (2.5)
Rash	2 (2.5)	2 (2.5)
Pneumonia	2 (2.5)	2 (2.5)
Myalgia	2 (2.5)	2 (2.5)

Relationship assessed by the investigator. Percentages are based on the number of subjects treated in each group. Overall N = 81.

Table 14.3.5 Serious Treatment-emergent Adverse Events by Preferred Term – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Anaemia	5 (6.2)	5 (6.2)
Malignant neoplasm progression	2 (2.5)	2 (2.5)
Ascites	1 (1.2)	1 (1.2)
Aspartate aminotransferase increased	1 (1.2)	1 (1.2)
Diarrhoea	1 (1.2)	1 (1.2)
Intestinal obstruction	1 (1.2)	1 (1.2)
Lymphopenia	1 (1.2)	1 (1.2)
Myelodysplastic syndrome	1 (1.2)	1 (1.2)
Pneumonia	1 (1.2)	1 (1.2)
Pneumonitis	1 (1.2)	1 (1.2)
Sepsis	1 (1.2)	1 (1.2)
Vomiting	1 (1.2)	1 (1.2)

Subjects are counted once per preferred term. Percentages are based on N = 81.

Table 14.3.6 Treatment-emergent Adverse Events Leading to Discontinuation of Study Treatment – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=81)	Total (N=81)
Malignant neoplasm progression	2 (2.5)	2 (2.5)
Ascites	1 (1.2)	1 (1.2)
Aspartate aminotransferase increased	1 (1.2)	1 (1.2)
Intestinal obstruction	1 (1.2)	1 (1.2)
Nausea	1 (1.2)	1 (1.2)
Pneumonia	1 (1.2)	1 (1.2)
Sepsis	1 (1.2)	1 (1.2)

Percentages are based on N = 81.

Table 14.3.7 Deaths – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=81)	Total (N=81)
Any death	12 (14.8)	12 (14.8)
Death within 30 days of last dose	5 (6.2)	5 (6.2)
Primary cause of death		
Disease progression	6 (7.4)	6 (7.4)
Adverse event	6 (7.4)	6 (7.4)
Other	2 (2.5)	2 (2.5)

Percentages are based on the number of subjects treated within each group.

Table 14.3.8 Shift from Baseline to Worst Post-baseline CTCAE Grade for Selected Laboratory Parameters – Safety Analysis Set

Laboratory parameter	Kelvaparib 300 mg: worst grade 3 n (%)	Kelvaparib 300 mg: worst grade 4 n (%)	Kelvaparib 300 mg: shift >= 2 grades n (%)	Total: worst grade 3 n (%)	Total: worst grade 4 n (%)	Total: shift >= 2 grades n (%)
Haemoglobin	12 (14.8)	3 (3.7)	27 (33.3)	12 (14.8)	3 (3.7)	27 (33.3)
Platelets	9 (11.1)	1 (1.2)	18 (22.2)	9 (11.1)	1 (1.2)	18 (22.2)
Neutrophils	4 (4.9)	0 (0.0)	14 (17.3)	4 (4.9)	0 (0.0)	14 (17.3)
Lymphocytes	6 (7.4)	1 (1.2)	17 (21.0)	6 (7.4)	1 (1.2)	17 (21.0)
CA-125	11 (13.6)	2 (2.5)	26 (32.1)	11 (13.6)	2 (2.5)	26 (32.1)
ALT	1 (1.2)	1 (1.2)	8 (9.9)	1 (1.2)	1 (1.2)	8 (9.9)
AST	0 (0.0)	3 (3.7)	10 (12.3)	0 (0.0)	3 (3.7)	10 (12.3)
Creatinine	4 (4.9)	0 (0.0)	9 (11.1)	4 (4.9)	0 (0.0)	9 (11.1)
Lactate dehydrogenase	9 (11.1)	0 (0.0)	13 (16.0)	9 (11.1)	0 (0.0)	13 (16.0)
Albumin	1 (1.2)	1 (1.2)	4 (4.9)	1 (1.2)	1 (1.2)	4 (4.9)

CTCAE = Common Terminology Criteria for Adverse Events, version 5.0. Percentages are based on the number of subjects treated in each group. Overall N = 81.

Table 14.3.9 Time to Onset and Resolution of the Most Frequent Treatment-emergent Adverse Events – Safety Analysis Set

Preferred term	Events, n	Median time to onset (days)	Resolved, n (%)	Median time to resolution (days)
Nausea	55	31.0	47 (85.5)	14.0
Fatigue	37	30.0	34 (91.9)	14.0
Anaemia	34	41.5	29 (85.3)	12.0
Vomiting	23	25.0	20 (87.0)	11.0
Constipation	21	47.0	19 (90.5)	6.0
Headache	20	37.0	18 (90.0)	7.0
Decreased appetite	19	12.0	17 (89.5)	10.0
Dysgeusia	18	30.0	16 (88.9)	7.0
Cough	17	11.0	17 (100.0)	12.0
Thrombocytopenia	17	23.0	12 (70.6)	12.0
Diarrhoea	14	9.5	13 (92.9)	7.0
Arthralgia	14	44.0	14 (100.0)	13.0
Neutropenia	14	37.0	10 (71.4)	13.0
Asthenia	13	41.0	13 (100.0)	7.0
Urinary tract infection	10	20.0	9 (90.0)	11.0
Insomnia	10	10.0	9 (90.0)	16.0
Abdominal pain	10	5.5	7 (70.0)	5.0
Oedema peripheral	10	35.0	10 (100.0)	11.5
Nasopharyngitis	10	18.0	10 (100.0)	5.0
Dizziness	9	31.0	8 (88.9)	2.0

Events are counted at the event level rather than the subject level. NE = not estimable.

14.3.10 Adverse events within subgroups

The summaries in Tables 14.3.10 to 14.3.16 are exploratory. The subgroups were not chosen or sized for comparisons of safety, the tables are not adjusted for multiplicity, and no conclusion about a difference between subgroups is drawn from them.

Table 14.3.10 Overview of adverse events by BRCA mutation status

Category	BRCA wild-type	BRCA1 mutated	BRCA2 mutated
Any event	52 (100.0)	19 (100.0)	10 (100.0)
Grade 3 or higher	23 (44.2)	5 (26.3)	2 (20.0)
Serious	15 (28.8)	1 (5.3)	1 (10.0)
Led to discontinuation	7 (13.5)	1 (5.3)	0 (0.0)
Fatal	6 (11.5)	0 (0.0)	0 (0.0)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.11 Overview of adverse events by homologous recombination status

Category	Negative	Positive
Any event	36 (100.0)	45 (100.0)
Grade 3 or higher	11 (30.6)	19 (42.2)
Serious	7 (19.4)	10 (22.2)
Led to discontinuation	4 (11.1)	4 (8.9)
Fatal	3 (8.3)	3 (6.7)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.12 Overview of adverse events by platinum sensitivity

Category	No	Yes
Any event	25 (100.0)	56 (100.0)
Grade 3 or higher	9 (36.0)	21 (37.5)
Serious	7 (28.0)	10 (17.9)
Led to discontinuation	5 (20.0)	3 (5.4)
Fatal	3 (12.0)	3 (5.4)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.13 Overview of adverse events by residual disease after surgery

Category	Absent	Present
Any event	48 (100.0)	33 (100.0)
Grade 3 or higher	20 (41.7)	10 (30.3)
Serious	12 (25.0)	5 (15.2)
Led to discontinuation	7 (14.6)	1 (3.0)
Fatal	6 (12.5)	0 (0.0)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.14 Overview of adverse events by prior lines of platinum-based therapy

Category	1 to 2	>= 3
Any event	56 (100.0)	25 (100.0)
Grade 3 or higher	18 (32.1)	12 (48.0)
Serious	11 (19.6)	6 (24.0)
Led to discontinuation	5 (8.9)	3 (12.0)
Fatal	4 (7.1)	2 (8.0)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.15 Overview of adverse events by age category

Category	<65	>=65
Any event	40 (100.0)	41 (100.0)
Grade 3 or higher	18 (45.0)	12 (29.3)
Serious	12 (30.0)	5 (12.2)
Led to discontinuation	5 (12.5)	3 (7.3)
Fatal	4 (10.0)	2 (4.9)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.16 Overview of adverse events by ECOG performance status

Category	0	1	2
Any event	26 (100.0)	47 (100.0)	8 (100.0)
Grade 3 or higher	11 (42.3)	18 (38.3)	1 (12.5)
Serious	6 (23.1)	11 (23.4)	0 (0.0)
Led to discontinuation	5 (19.2)	3 (6.4)	0 (0.0)
Fatal	4 (15.4)	2 (4.3)	0 (0.0)

Percentages are calculated on the number of subjects in that subgroup.

14.4 Pharmacokinetic data

Table 14.4.1 Summary of Steady-state Pharmacokinetic Parameters – Pharmacokinetic Analysis Set

Treatment	n	Cmax (ng/mL) Geometric mean (%CV)	tmax (h) Median (min, max)	AUC0-12 (ng·h/mL) Geometric mean (%CV)	t1/2 (h) Mean (SD)	CL/F (L/h) Geometric mean (%CV)
Kelvaparib 300 mg	81	1757 (28.0)	2.0 (2.0, 2.0)	17671 (32.5)	14.5 (6.46)	16.98 (32.5)

AUC0-12 = area under the concentration-time curve over one dosing interval; CL/F = apparent oral clearance; Cmax = maximum observed concentration; %CV = geometric coefficient of variation; t1/2 = terminal elimination half-life; tmax = time to maximum concentration.

Table 14.4.2 Accumulation Ratio and Trough Concentration – Pharmacokinetic Analysis Set

Treatment	n	Accumulation ratio Geometric mean (%CV)	Ctrough (ng/mL) Geometric mean (%CV)
Kelvaparib 300 mg	81	2.86 (24.6)	1094.6 (42.8)

Accumulation ratio is the ratio of AUC0-24 at steady state to AUC0-24 after the first dose.

15 REFERENCE LIST

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16 APPENDICES

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